

# Skinterest Tech

## Multi-Modal Skin Condition Classification

AI Studio Final Project

12/6/2025

**SKINTEREST**

**BREAK  
THROUGH  
TECH**

# Meet Our Team



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# Our AI Studio Coach & Challenge Advisor



**Thandiwe-Kesi Robins**  
Challenge Advisor

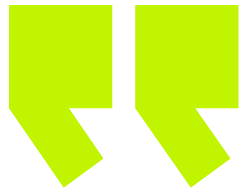


**Ashley Abid**  
Challenge Advisor



**Nandini Proothi**  
Coach

**SKINTEREST**



*Only **17.8% of educational social media posts**  
represented darker skin tones.*

*Only **4.5% of images in medical textbooks**  
represented darker skin tones*

Skin conditions may not look the same on every skin tone and lighting. How can we build a skin condition classification application that works for everyone when the data is skewed?



# Our Goals

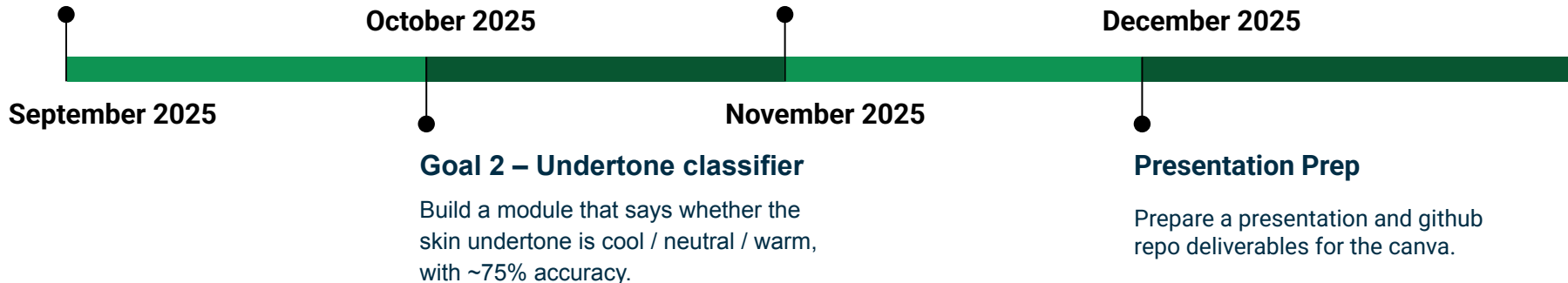
We are building an ML system that can classify images of skin conditions with a diverse and imperfect set of data

## Goal 1 – Lighting / color calibration

Build a model that can tell harsh vs soft lighting and help recover a more “true” view of the skin, with about 80% accuracy.

## Goal 3 – Put it together + test

Check that these pieces still work and stay stable when you test them on a separate, mixed set of eczema / rosacea / acne images.



# Resources We Leveraged



**Hugging Face**

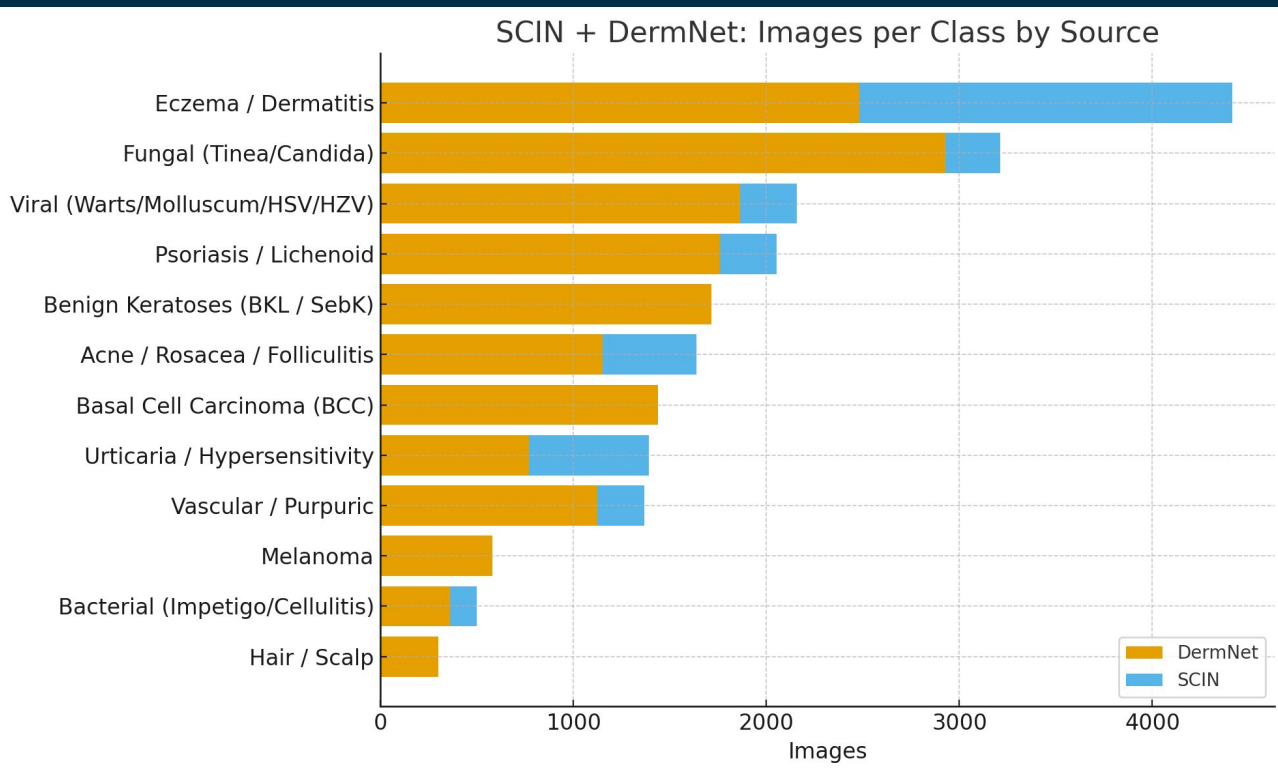


**slack**



1. **GitHub Board:** tracked milestones, experiments, and dataset issues
2. **Google Colab:** used for training, fine-tuning, Grad-CAM testing
3. **Slack:** collaboration, sharing results, model comparisons
4. **HuggingFace Hub:** used for sharing large models (>100MB)
5. **VS Code + Streamlit:** local environment for our interactive prototype

# Data Snapshot and Overview

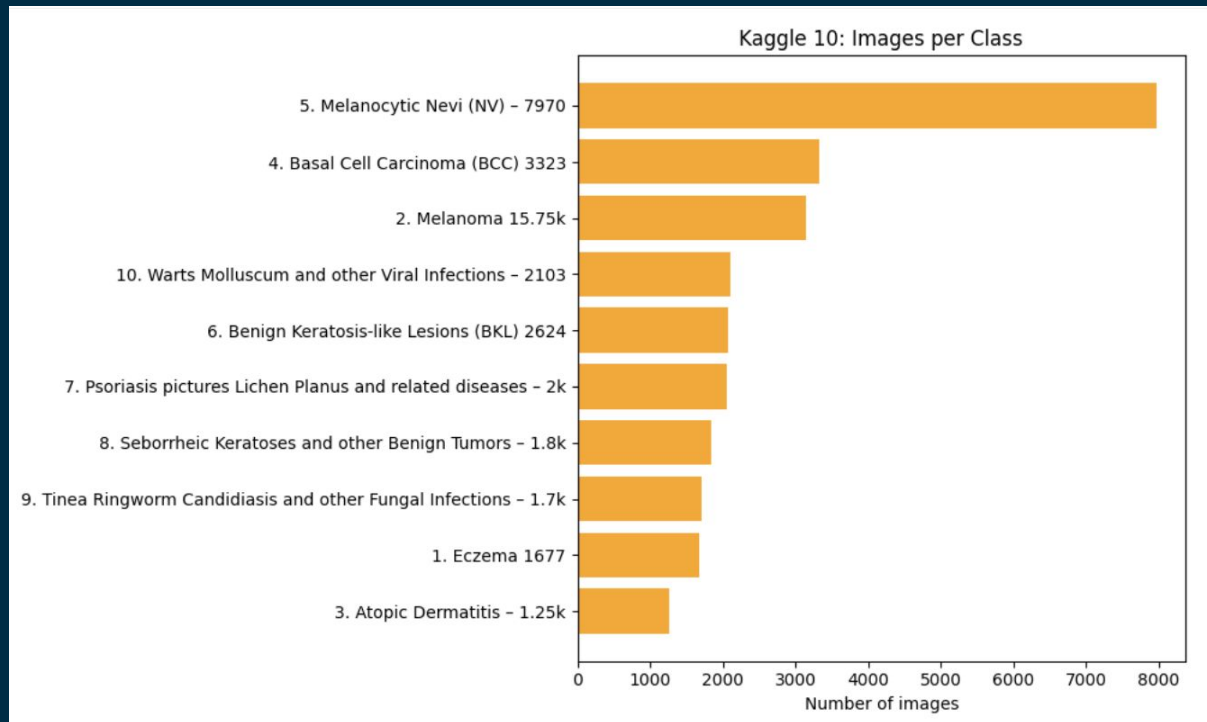


*DermNet/Kaggle:* curated, clinic-style photos from an online medical education site (~19.5k images across 23 diseases; consistent views but mostly lighter tones and “ideal” shots).

*SCIN:* real-world, volunteer “phone photos” from US Google users (10k+ images with self-reported demographics and skin tone, dermatologist labels, etc.).

**Graph 1.** Scin + Dermnet Image Numbers per Class.

# Data Snapshot and Overview

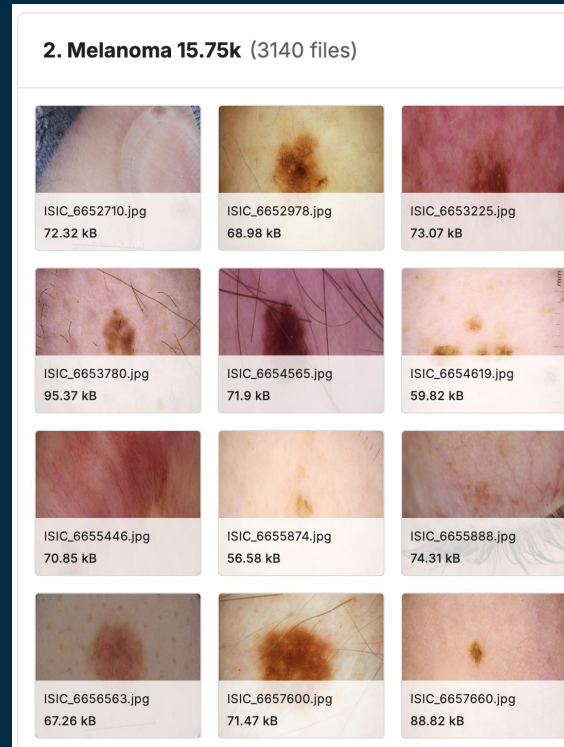
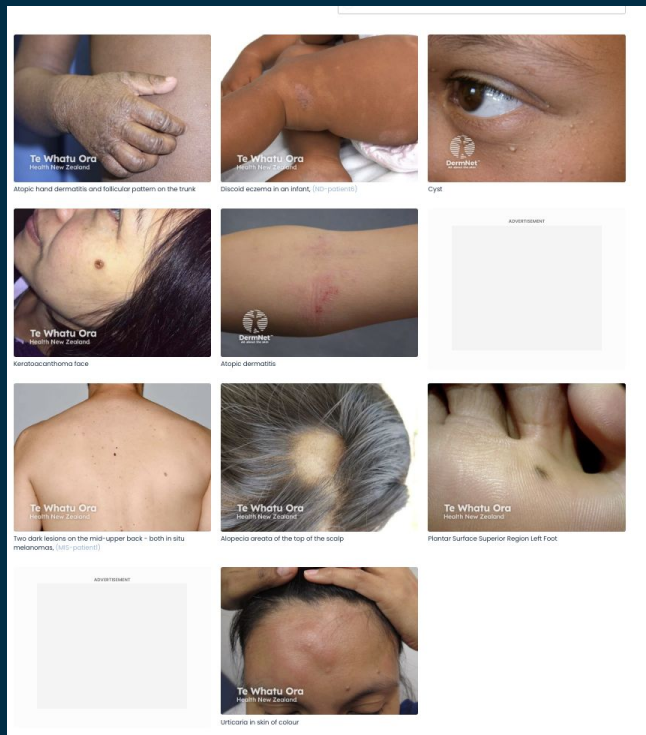


Graph 2. Kaggle's Skin Disease Dataset Image Numbers per Class.

## Kaggle Skin Disease Dataset (27,000 Images, 10 Classes)

- High-resolution dermatology photos from clinical sources
- Limitations: low diversity, mostly light skin tones, highly zoomed clinical photos
- Used as our **baseline model** before adding more diverse datasets (SCIN, DermNet)

# Data Snapshot and overview



Dermnet

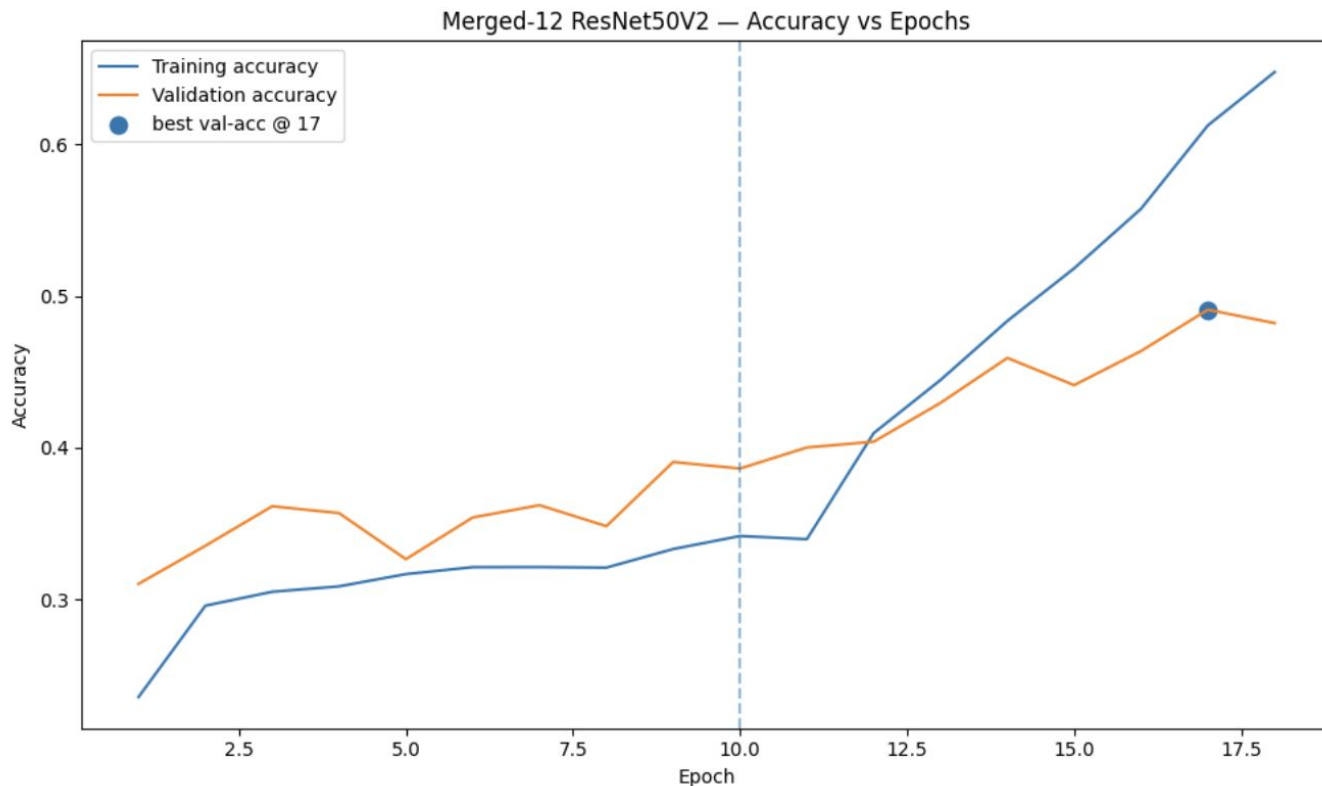
SCIN

Skin diseases image dataset  
(kaggle)

# Data Preprocessing

- Unified 34 SCIN labels → 12 canonical classes
- Kept only dermatologist-labeled, gradable SCIN images
- Removed missing/multi-condition labels & duplicates
- Cropped, resized, normalized & lightly augmented all images
- Ensured patient-level train/val/test splits

# Model Selection Process



Tested **Xception**,  
**ResNet50V2**,  
**EfficientNetB0**, **MobileNet**,  
and hybrid custom CNNs

Evaluated models using  
accuracy, confusion matrices,  
per-class recall, and  
generalization to SCIN  
images

Chose an **ensemble  
approach** because no single  
model performed best across  
all skin tones & lighting

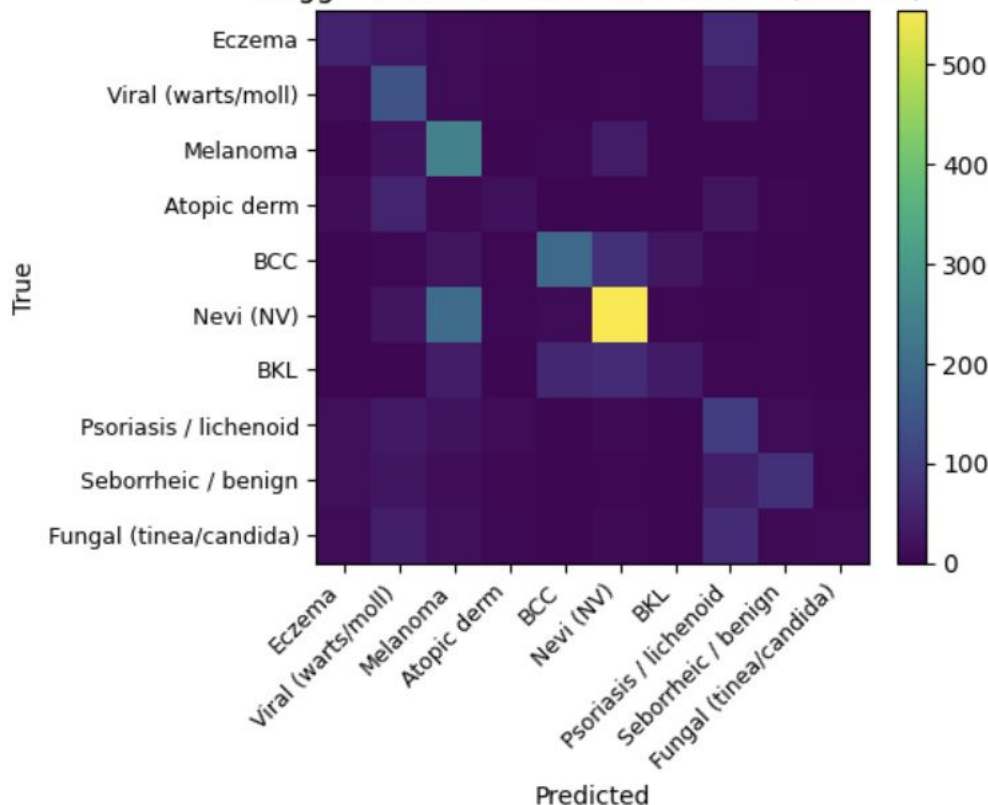
Added a **Lighting/Undertone  
ResNet head** to assess  
photo quality before  
classification

# Model Comparison

| Model Name                        | Description                                    | Results                                          | Pros                                                 | Cons                                     |
|-----------------------------------|------------------------------------------------|--------------------------------------------------|------------------------------------------------------|------------------------------------------|
| Kaggle 10-class ResNetV2 baseline | ResNetV2 on 10 clean Kaggle classes.           | Top-1 $\approx 0.53$ ,<br>Top-3 $\approx 0.83$ . | Simple, fast, strong baseline.                       | Less diverse, no fairness checks.        |
| Xception + ResNet50V2 ensemble    | Two-model ensemble on merged 12-class dataset. | Top-1 $\approx 0.55$ ,<br>Top-3 $\approx 0.83$ . | Best macro performance, more robust across diseases. | Heavier, still confuses similar classes. |
| EffNetB0 + Custom CNN ensemble    | Lighter ensemble on same 12 classes.           | Top-1 $\approx 0.53$ ,<br>Top-3 $\approx 0.82$ . | More efficient; good for deployment.                 | Slightly weaker than Xception+ResNet.    |

# Kaggle 10-Class Model Results

Kaggle 10-class - Confusion matrix (test set)

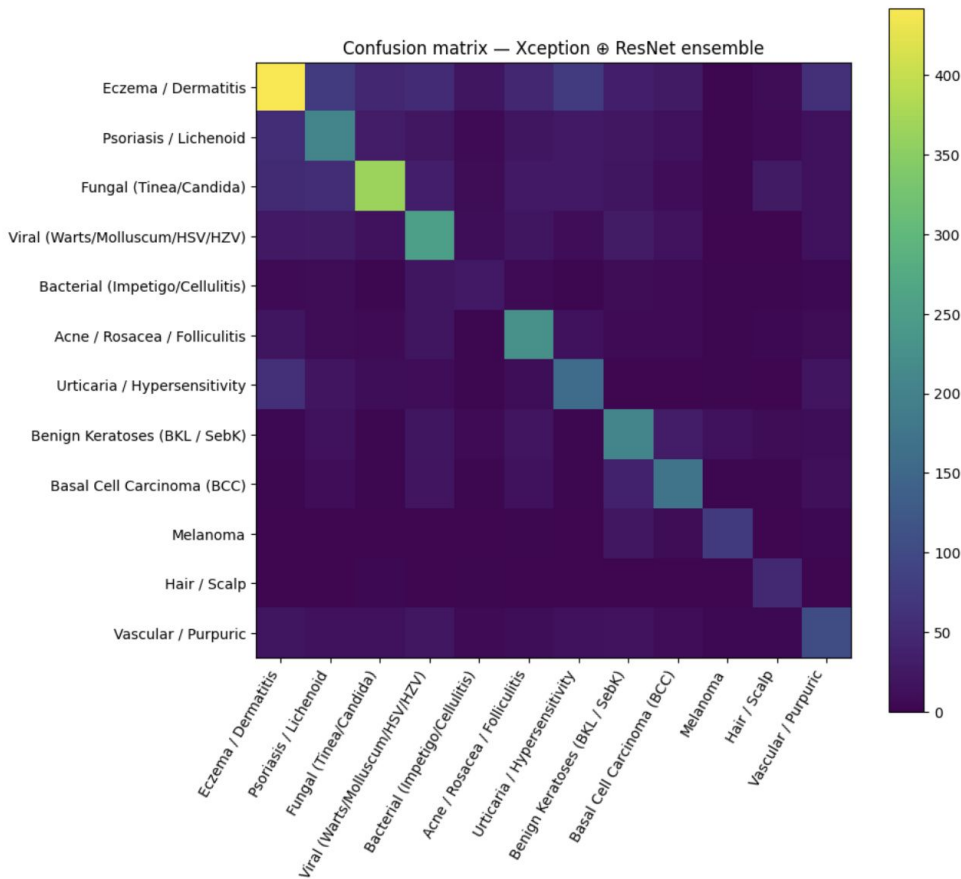


Our **Top-1 accuracy is around 52%**, and **Top-3 is about 82%**, which is decent given the strong class imbalance.

|                                                                  | precision | recall | f1-score | support |
|------------------------------------------------------------------|-----------|--------|----------|---------|
| 1. Ecema 1677                                                    | 0.42      | 0.30   | 0.35     | 167     |
| 10. Warts Molluscum and other Viral Infections - 2103            | 0.38      | 0.68   | 0.48     | 211     |
| 2. Melanoma 15.75k                                               | 0.42      | 0.79   | 0.55     | 314     |
| 3. Atopic Dermatitis - 1.25k                                     | 0.32      | 0.15   | 0.21     | 125     |
| 4. Basal Cell Carcinoma (BCC) 3323                               | 0.70      | 0.57   | 0.63     | 333     |
| 5. Melanocytic Nevi (NV) - 7970                                  | 0.73      | 0.70   | 0.71     | 797     |
| 6. Benign Keratosis-like Lesions (BKL) 2624                      | 0.51      | 0.17   | 0.26     | 208     |
| 7. Psoriasis pictures Lichen Planus and related diseases - 2k    | 0.30      | 0.48   | 0.37     | 206     |
| 8. Seborrheic Keratoses and other Benign Tumors - 1.8k           | 0.70      | 0.40   | 0.51     | 185     |
| 9. Tinea Ringworm Candidiasis and other Fungal Infections - 1.7k | 0.42      | 0.06   | 0.11     | 170     |
| accuracy                                                         |           |        | 0.53     | 2716    |
| macro avg                                                        | 0.49      | 0.43   | 0.42     | 2716    |
| weighted avg                                                     | 0.56      | 0.53   | 0.51     | 2716    |

Top-1: 0.525  
Top-3: 0.825

# Xception + ResNet50V2 Ensemble Results



Top-1: 0.546

Top-3: 0.826

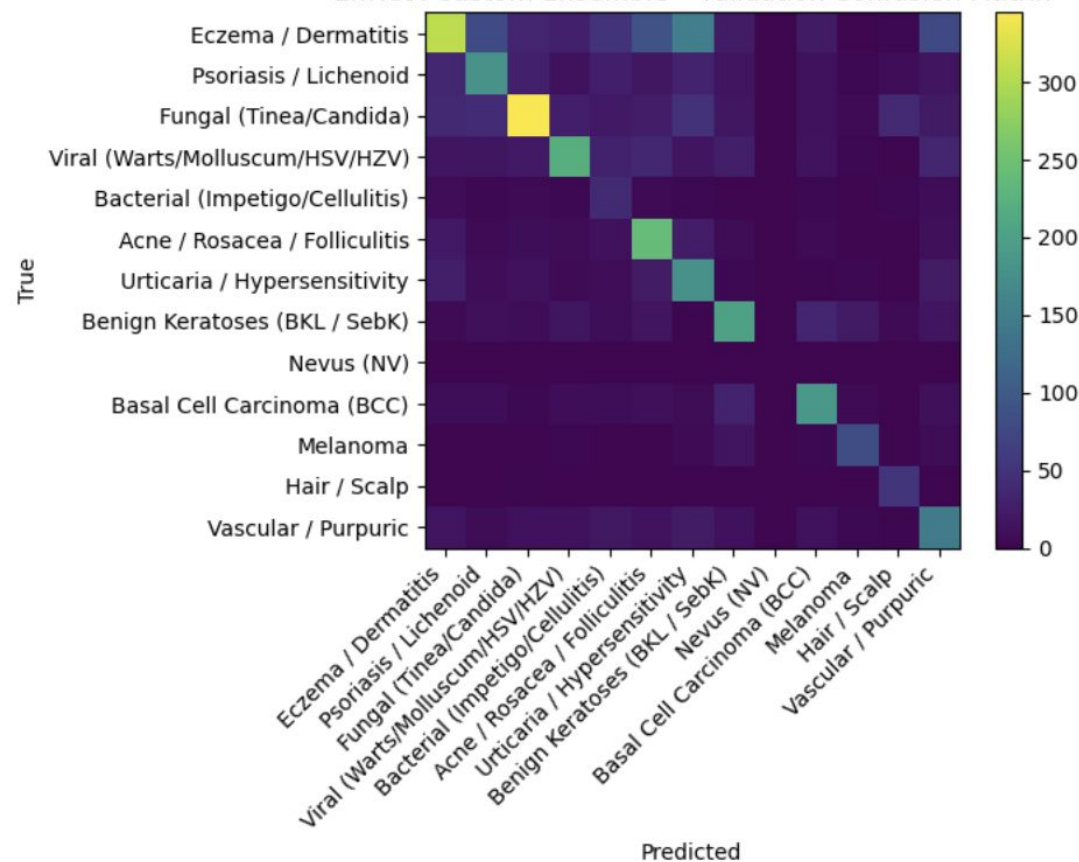
|                                 | precision | recall | f1-score | support |
|---------------------------------|-----------|--------|----------|---------|
| Eczema / Dermatitis             | 0.638     | 0.492  | 0.555    | 899     |
| Psoriasis / Lichenoid           | 0.461     | 0.473  | 0.467    | 433     |
| Fungal (Tinea/Candida)          | 0.723     | 0.564  | 0.634    | 647     |
| Viral (Warts/Molluscum/HSV/HZV) | 0.522     | 0.592  | 0.555    | 429     |
| Bacterial (Impetigo/Cellulitis) | 0.287     | 0.284  | 0.286    | 95      |
| Acne / Rosacea / Folliculitis   | 0.555     | 0.696  | 0.618    | 326     |
| Urticaria / Hypersensitivity    | 0.469     | 0.541  | 0.502    | 292     |
| Benign Keratoses (BKL / SebK)   | 0.504     | 0.603  | 0.549    | 343     |
| Basal Cell Carcinoma (BCC)      | 0.558     | 0.603  | 0.580    | 287     |
| Melanoma                        | 0.673     | 0.627  | 0.649    | 118     |
| Hair / Scalp                    | 0.438     | 0.817  | 0.570    | 60      |
| Vascular / Purpuric             | 0.381     | 0.406  | 0.393    | 254     |
| accuracy                        |           |        | 0.546    | 4183    |
| macro avg                       | 0.517     | 0.558  | 0.530    | 4183    |
| weighted avg                    | 0.561     | 0.546  | 0.547    | 4183    |

The ensemble's **Top-1 accuracy increases to ~54–55%**, and **Top-3 stays high at ~82–83%**.

The confusion matrix shows cleaner diagonal structure, meaning fewer cross-class mistakes.

# EfficientNetB0 + Custom CNN Ensemble Results

EffNet+Custom Ensemble - Validation Confusion Matrix



EffNet+Custom ensemble - Top-1: 0.528

EffNet+Custom ensemble - Top-3: 0.822

|                                 | precision | recall | f1-score | support |
|---------------------------------|-----------|--------|----------|---------|
| Eczema / Dermatitis             | 0.640     | 0.356  | 0.458    | 862     |
| Psoriasis / Lichenoid           | 0.497     | 0.459  | 0.477    | 388     |
| Fungal (Tinea/Candida)          | 0.720     | 0.534  | 0.613    | 646     |
| Viral (Warts/Molluscum/HSV/HZV) | 0.638     | 0.509  | 0.566    | 436     |
| Bacterial (Impetigo/Cellulitis) | 0.180     | 0.471  | 0.260    | 87      |
| Acne / Rosacea / Folliculitis   | 0.505     | 0.711  | 0.591    | 339     |
| Urticaria / Hypersensitivity    | 0.362     | 0.606  | 0.453    | 292     |
| Benign Keratoses (BKL / SebK)   | 0.574     | 0.587  | 0.580    | 344     |
| Nevus (NV)                      | 0.000     | 0.000  | 0.000    | 0       |
| Basal Cell Carcinoma (BCC)      | 0.608     | 0.657  | 0.632    | 286     |
| Melanoma                        | 0.633     | 0.698  | 0.664    | 116     |
| Hair / Scalp                    | 0.444     | 0.881  | 0.591    | 59      |
| Vascular / Purpuric             | 0.400     | 0.529  | 0.455    | 280     |
| accuracy                        |           |        | 0.528    | 4135    |
| macro avg                       | 0.477     | 0.538  | 0.488    | 4135    |
| weighted avg                    | 0.572     | 0.528  | 0.531    | 4135    |

Its **Top-1 accuracy is about 52–53%**, very close to the Xception+ResNet ensemble.

Top-3 accuracy is around **82%**.

Interestingly, this model generalizes slightly better on **skin-tone subgroups**, which was one of our project goals.

# What We Learned

- **Data quality & diversity really matter.** Clean Kaggle images gave solid baseline accuracy, but adding SCIN + DermNet showed how more realistic, messy photos change performance and highlight gaps for darker skin tones.
- **Architectures trade off strength on different classes.** Single models (Kaggle, Xception, EffNet) each had blind spots; ensembles smoothed those out and gave more stable per-class performance instead of just chasing overall accuracy.
- **Fairness needs its own head.** Building the lighting + undertone model taught us that “is this photo even usable?” and “does it work across tones?” are separate questions, and you need explicit checks for both, not just a big disease classifier

# Streamlit demo

<https://aishasalim-skinterest-2b-app-56ogvh.streamlit.app/>



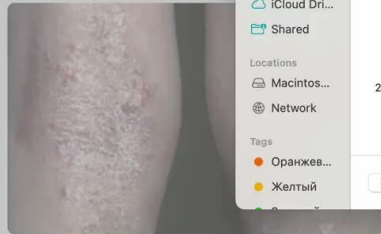
## Skin photo classifier — Kaggle • Merged-12 Ensembles

Prototype only — not medical advice.

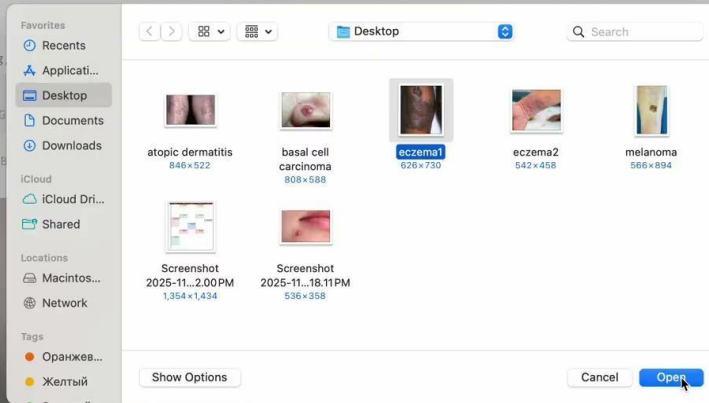
Upload a phone-style skin photo (jpg / jpeg)

Drag and drop file here  
Limit 200MB per file • JPG, JPEG

atopic dermatitis.png 0.8MB



Uploaded image



Kaggle Skin Diseases Dataset based model (10 diseases)

Ensemble of Xception + ResNet (12 categories)

Ensemble of EffNetB0 + Custom CNN (12 categories)

< Manage app

Thank you!

# Questions?